

0.1 重积分不等式证明

例题 0.1

(1) 定义在 $[0, 1] \times [0, 1]$ 的函数 $f(x, y)$ 连续可微, 且在边界上恒为 0, 求证:

$$\left| \iint_{[0,1] \times [0,1]} f(x, y) dx dy \right| \leq \frac{1}{6} \max_{(x,y) \in [0,1] \times [0,1]} \sqrt{\left(\frac{\partial f}{\partial x}\right)^2 + \left(\frac{\partial f}{\partial y}\right)^2}.$$

(2) 设 $D = \{(x, y) : x^2 + y^2 \leq R^2\}$, 函数 f 在 D 上连续可微, 且在边界 ∂D 上恒为 0, 证明:

$$\iint_D f(x, y) dx dy \leq \frac{\pi R^3}{3} \max_{(x,y) \in D} \sqrt{f_x^2 + f_y^2}.$$

证明 Show/Hide Proof

(1) 用直线 $y = x$ 与 $y = 1 - x$ 将正方形 $[0, 1]^2$ 分成四个全等的三角形区域, 记为 D_1, D_2, D_3, D_4 , 其中

$$D_1 = \{(x, y) : \frac{1}{2} \leq y \leq 1, 1 - y \leq x \leq y\},$$

$$D_2 = \{(x, y) : 0 \leq y \leq \frac{1}{2}, y \leq x \leq 1 - y\},$$

$$D_3 = \{(x, y) : 0 \leq x \leq \frac{1}{2}, x \leq y \leq 1 - x\},$$

$$D_4 = \{(x, y) : \frac{1}{2} \leq x \leq 1, 1 - x \leq y \leq x\}.$$

令 $M \triangleq \max_{[0,1]^2} \sqrt{f_x^2 + f_y^2}$. 对 $\forall (x, y) \in D_1$, 固定 x , 因为函数 $y \mapsto f(x, y)$ 在 $[0, 1]$ 上可微且 $f(x, 1) = 0$, 所以由 Lagrange 中值定理, 存在 $\eta \in (y, 1)$ 使得

$$|f(x, y)| = |f(x, y) - f(x, 1)| = |(y - 1)f_y(x, \eta)| \leq (1 - y)|f_y(x, \eta)| \leq (1 - y)M.$$

因此

$$\begin{aligned} \iint_{D_1} |f(x, y)| dx dy &\leq M \iint_{D_1} (1 - y) dx dy = M \int_{\frac{1}{2}}^1 (1 - y) \left(\int_{1-y}^y dx \right) dy \\ &= M \int_{\frac{1}{2}}^1 (1 - y)(2y - 1) dy = M \left[-\frac{2}{3}y^3 + \frac{3}{2}y^2 - y \right]_{\frac{1}{2}}^1 = \frac{M}{24}. \end{aligned}$$

同理, 对其余三个三角形区域 D_2, D_3, D_4 , 分别利用 $f(x, 0) = 0$, $f(0, y) = 0$, $f(1, y) = 0$ 作类似的估计可得

$$\iint_{D_k} |f(x, y)| dx dy \leq \frac{M}{24}, \quad k = 2, 3, 4.$$

于是

$$\iint_{[0,1]^2} f(x, y) dx dy \leq \iint_{[0,1]^2} |f(x, y)| dx dy = \sum_{k=1}^4 \iint_{D_k} |f(x, y)| dx dy \leq 4 \cdot \frac{M}{24} = \frac{M}{6}.$$

(2) 令

$$M \triangleq \max_{(x,y) \in D} \sqrt{f_x^2 + f_y^2}.$$

做极坐标变换 $x = r \cos \theta$, $y = r \sin \theta$, 其中 $0 \leq r \leq R$, $0 \leq \theta \leq 2\pi$. 边界条件变为 $f(R, \theta) = 0$.

对 $\forall (r, \theta) \in [0, R] \times [0, 2\pi]$, 固定 θ , 因为函数 $r \mapsto f(r, \theta)$ 在 $[0, R]$ 上连续可微且 $f(R, \theta) = 0$, 所以由 Lagrange 中值定理, 存在 $\xi \in (r, R)$, 使得

$$|f(r, \theta)| = |f(r, \theta) - f(R, \theta)| = |(r - R)f_r(\xi, \theta)| \leq (R - r)|f_r(\xi, \theta)|.$$

由于

$$|f_r| = |f_x \cos \theta + f_y \sin \theta| \leq \sqrt{f_x^2 + f_y^2} \leq M,$$

故

$$|f(r, \theta)| \leq M(R - r).$$

于是

$$\iint_D |f(x, y)| \, dx dy = \int_0^{2\pi} \int_0^R |f(r, \theta)| r \, dr d\theta \leq M \int_0^{2\pi} \int_0^R (R-r) r \, dr d\theta = M \cdot 2\pi \cdot \frac{R^3}{6} = \frac{\pi R^3}{3} M.$$

□

例题 0.2

(1) 设 $u \in C^1[0, 1]$ 满足 $u(0) = 0$, 证明:

$$|u(1)|^2 \leq \tanh 1 \cdot \left(\int_0^1 [|u(r)|^2 + |u'(r)|^2] \, dr \right).$$

(2) 设 $B \subset \mathbb{R}^3$ 是单位球, 证明:

$$\int_{\partial B} |f|^2 \, dS \leq \frac{e^2 - 1}{2} \left(\int_B |f|^2 \, dV + \int_B |\nabla f|^2 \, dV \right), \quad \forall f \in C^\infty(\bar{B}),$$

并证明系数不可改进.

注

(1) 第 (1) 问中, 为了分部积分能表示出 $u(1)$, 我们只需令 η 满足 $\eta'' = \eta, \eta'(1) = 1$.

(2) 第 (2) 问的思路是: 用极坐标将对球体的积分转化为径向部分的函数的积分.

证明 [Show/Hide Proof](#)

(1) 令 $\eta(r) \triangleq \frac{\sinh r}{\cosh 1}$, 则

$$\eta'(r) = \frac{\cosh r}{\cosh 1}, \quad \eta''(r) = \frac{\sinh r}{\cosh 1} = \eta(r).$$

由分部积分知

$$\int_0^1 u'(r) \eta'(r) \, dr = u(1) \eta'(1) - \int_0^1 u(r) \eta''(r) \, dr = u(1) - \int_0^1 u(r) \eta(r) \, dr.$$

从而

$$u(1) = \int_0^1 (u'(r) \eta'(r) + u(r) \eta(r)) \, dr.$$

于是由 Cauchy 不等式和命题??可得

$$\begin{aligned} |u(1)|^2 &\leq \left| \int_0^1 (u'(r) \eta'(r) + u(r) \eta(r)) \, dr \right|^2 \\ &\stackrel{\text{Cauchy 不等式 (离散版)}}{\leq} \left| \int_0^1 \sqrt{|u'(r)|^2 + |u(r)|^2} \cdot \sqrt{|\eta'(r)|^2 + |\eta(r)|^2} \, dr \right|^2 \\ &\stackrel{\text{Cauchy 积分不等式}}{\leq} \left(\int_0^1 (|u'(r)|^2 + |u(r)|^2) \, dr \right) \left(\int_0^1 (|\eta'(r)|^2 + |\eta(r)|^2) \, dr \right) \\ &= \left(\int_0^1 (|u'(r)|^2 + |u(r)|^2) \, dr \right) \int_0^1 \frac{\cosh^2 r + \sinh^2 r}{\cosh^2 1} \, dr \\ &= \left(\int_0^1 (|u'(r)|^2 + |u(r)|^2) \, dr \right) \int_0^1 \frac{\cosh(2r)}{\cosh^2 1} \, dr \\ &= \left(\int_0^1 (|u'(r)|^2 + |u(r)|^2) \, dr \right) \cdot \frac{\sinh 2}{2 \cosh^2 1} \\ &= \tanh 1 \cdot \int_0^1 (|u'(r)|^2 + |u(r)|^2) \, dr. \end{aligned}$$

(2) 对 $\forall w > 0$, 令 $F_w(r) \triangleq r f(rw)$ ($\forall r \in [0, 1]$), 则利用 (1) 可得

$$\begin{aligned} |f(w)|^2 &= |F_w(1)|^2 \leq \tanh 1 \int_0^1 (|F_w(r)|^2 + |F'_w(r)|^2) \, dr \\ &= \tanh 1 \int_0^1 \left(r^2 |f(rw)|^2 + \left| f(rw) + r \frac{\partial f(rw)}{\partial r} \right|^2 \right) \, dr \end{aligned}$$

$$\begin{aligned}
&= \tanh 1 \int_0^1 \left[r^2 |f(rw)|^2 + r^2 \left| \frac{\partial f(rw)}{\partial r} \right|^2 + |f(rw)|^2 + 2r^2 f(rw) \frac{\partial f(rw)}{\partial r} \right] dr \\
&= \tanh 1 \int_0^1 \left[r^2 |f(rw)|^2 + r^2 \left| \frac{\partial f(rw)}{\partial r} \right|^2 + \frac{d(r|f(rw)|^2)}{dr} \right] dr \\
&= \tanh 1 \int_0^1 \left(r^2 |f(rw)|^2 + r^2 \left| \frac{\partial f(rw)}{\partial r} \right|^2 \right) dr + \tanh 1 |f(w)|^2.
\end{aligned}$$

从而

$$\begin{aligned}
|f(w)| &\leq \frac{\tanh 1}{1 - \tanh 1} \int_0^1 \left(r^2 |f(rw)|^2 + r^2 \left| \frac{\partial f(rw)}{\partial r} \right|^2 \right) dr \\
&= \frac{e^2 - 1}{2} \int_0^1 \left(r^2 |f(rw)|^2 + r^2 \left| \frac{\partial f(rw)}{\partial r} \right|^2 \right) dr.
\end{aligned}$$

于是

$$\begin{aligned}
\int_{\partial B} |f(w)|^2 dS(w) &\leq \frac{e^2 - 1}{2} \int_{\partial B} \int_0^1 \left(r^2 |f(rw)|^2 + r^2 \left| \frac{\partial f(rw)}{\partial r} \right|^2 \right) dr dS(w) \\
&= \frac{e^2 - 1}{2} \int_0^1 \int_{|w|=1} \left(r^2 |f(rw)|^2 + r^2 |w \cdot \nabla f(rw)|^2 \right) dS(w) dr \\
&\stackrel{\text{第一类曲面积分换元法}}{=} \frac{e^2 - 1}{2} \int_0^1 \int_{|v|=r} \left(|f(v)|^2 + \left| \frac{v}{r} \cdot \nabla f(v) \right|^2 \right) dS(v) dr \\
&= \frac{e^2 - 1}{2} \int_0^1 \int_{|v|=r} \left(|f(v)|^2 + |\nabla f(v)|^2 \right) dS(v) dr \\
&\stackrel{\text{余面积公式}}{=} \frac{e^2 - 1}{2} \iiint_{0 \leq |v| \leq 1} \left(|f(v)|^2 + |\nabla f(v)|^2 \right) dV \\
&= \frac{e^2 - 1}{2} \int_B \left(|f|^2 + |\nabla f|^2 \right) dV.
\end{aligned}$$

特别地, 当

$$f(x, y, z) = \frac{\sinh \sqrt{x^2 + y^2 + z^2}}{\sqrt{x^2 + y^2 + z^2}}$$

时, 不等式等号成立.

□